

Part d Verification of FastAPI - Improve the documentation (using errors)

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ uvicorn main:api --reload
INFO: Will watch for changes in these directories: ['/home/ubuntu/datascientest-data-engineering/fast-api']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [1912] using StatReload
INFO: Started server process [1917]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Edit [main.py](#) file:

```
main.py M X
datascientest-data-engineering > fast-api >
1  from fastapi import FastAPI # type: ignore
2
3
4  api = FastAPI()
5  data = [1, 2, 3, 4, 5]
6
7  @api.get("/data")
8  def get_data(index):
9      return {
10         "data": data[int(index)]
11     }
```

Open URL <http://127.0.0.1:8000/docs>

FastAPI 0.1.0 OAS 3.1
/openapi.json

default

GET /data Get Data

Schemas

HTTPValidationError > Expand all object

ValidationError > Expand all object

Run following commands in new terminal:

```
ubuntu@ip-172-31-18-113:~$ cd datascientest-data-engineering/fast-api/
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i http://127.0.0.1:8000/data?
index=my_index
HTTP/1.1 500 Internal Server Error
date: Wed, 18 Jun 2025 14:15:16 GMT
server: uvicorn
content-length: 21
content-type: text/plain; charset=utf-8

Internal Server Errorubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i http://
/127.0.0.1:8000/data?index=1234
HTTP/1.1 500 Internal Server Error
date: Wed, 18 Jun 2025 14:15:33 GMT
server: uvicorn
content-length: 21
content-type: text/plain; charset=utf-8

Internal Server Errorubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$
```

Open URL http://127.0.0.1:8000/data?index=my_index

Internal Server Error

Open URL <http://127.0.0.1:8000/data?index=1234>

Internal Server Error

Edit [main.py](#) file again:

```
main.py M X
datascientest-data-engineering > fast-api > main.py > .
1  from fastapi import FastAPI, HTTPException
2
3
4  api = FastAPI()
5  data = [1, 2, 3, 4, 5]
6
7  @api.get("/data")
8  def get_data(index):
9      try:
10         return {
11             "data": data[int(index)]
12         }
13     except IndexError:
14         raise HTTPException(
15             status_code = 404,
16             detail = "Unknown Index"
17         )
18     except ValueError:
19         raise HTTPException(
20             status_code = 400,
21             detail = "Bad Type"
22         )
```

Run commands in separate terminal:

```
ubuntu@ip-172-31-18-113:~$ cd datascientest-data-engineering/fast-api/
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i "http://127.0.0.1:8000/data?index=my_index"
HTTP/1.1 400 Bad Request
date: Wed, 18 Jun 2025 14:41:15 GMT
server: uvicorn
content-length: 21
content-type: application/json

{"detail": "Bad Type"}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i "http://127.0.0.1:8000/data?index=1234"
HTTP/1.1 404 Not Found
date: Wed, 18 Jun 2025 14:41:25 GMT
server: uvicorn
content-length: 26
content-type: application/json

{"detail": "Unknown Index"}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$
```

Open URL <http://127.0.0.1:8000/docs>

FastAPI 0.100 OAS 3.1

/openapi.json

default

GET /data Get Data

Schemas

HTTPValidationError > Expand all object

ValidationError > Expand all object

Open URL http://127.0.0.1:8000/data?index=my_index

```
{
  "detail": "Bad Type"
}
```

Raw Parsed

Open URL <http://127.0.0.1:8000/data?index=1234>

```
{
  "detail": "Unknown Index"
}
```

Raw Parsed

Edit [main.py](#) file again:

```
main.py M X
datascientest-data-engineering > fast-api > main.py > MyException
1 from fastapi import FastAPI, Request # type: ignore
2 from fastapi.responses import JSONResponse # type: ignore
3 import datetime
4
5
6 api = FastAPI()
7
8 class MyException(Exception):
9     def __init__(self, name: str, date: str):
10         self.name = name
11         self.date = date
12
13 @api.exception_handler(MyException)
14 def MyExceptionHandler(request: Request, exception: MyException):
15     return JSONResponse(
16         status_code = 418,
17         content = {
18             "url": str(request.url),
19             "name": exception.name,
20             "message": "This error is my own",
21             "date": exception.date
22         }
23     )
24
25 @api (function) def get_my_custom_exception() -> NoReturn
26 def get_my_custom_exception():
27     raise MyException(
28         name = "my error",
29         date = str(datetime.datetime.now())
30     )
```

Run command in separate terminal:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i http://127.0.0.1:8000/my_custom_exception
HTTP/1.1 418
date: Wed, 18 Jun 2025 15:00:13 GMT
server: uvicorn
content-length: 138
content-type: application/json

{"url": "http://127.0.0.1:8000/my_custom_exception", "name": "my error", "message": "This error is my own", "date": "2025-06-18 15:00:13.112811"}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$
```

Open URL http://127.0.0.1:8000/my_custom_exception

```
{
  "url": "http://127.0.0.1:8000/my_custom_exception",
  "name": "my error",
  "message": "This error is my own",
  "date": "2025-06-18 15:00:32.111489"
}
```

Raw Parsed

Edit [main.py](#) file one last time:

```
main.py M X
datascientest-data-engineering > fast-api > main.py > ...
1 from fastapi import FastAPI, Request # type: ignore
2 from fastapi.responses import JSONResponse # type: ignore
3 import datetime
4
5
6 api = FastAPI()
7
8 class MyException(Exception):
9     def __init__(self, name: str, date: str):
10         self.name = name
11         self.date = date
12
13 @api.exception_handler(MyException)
14 def MyExceptionHandler(request: Request, exception: MyException):
15     return JSONResponse(
16         status_code=418,
17         content={
18             "url": str(request.url),
19             "name": exception.name,
20             "message": "This error is my own",
21             "date": exception.date
22         }
23     )
24
25 @api.get("/my_custom_exception")
26 def get_my_custom_exception():
27     raise MyException(
28         name="my error",
29         date=str(datetime.datetime.now())
30     )
31
32 responses = {
33     200: {"description": "OK"},
34     404: {"description": "Item not found"},
35     302: {"description": "The item was moved"},
36     403: {"description": "Not enough privileges"},
37 }
38
39 @api.get("/thing", responses=responses)
40 def get_thing():
41     return {"data": "hello world"}
```

Open URL <http://127.0.0.1:8000/docs>

FastAPI

0.1.0 OAS 3.1

/openapi.json

default

GET /my_custom_exception Get My Custom Exception

GET /thing Get Thing

Parameters

No parameters

Responses

Code	Description	Links
200	OK Media type application/json Controls Accept header. Example Value Schema "string"	No links
302	The item was moved	No links
403	Not enough privileges	No links
404	Item not found	No links

Open URL <http://127.0.0.1:8000/redoc>

Search...

GET Get My Custom Exception

GET Get Thing

Get Thing

Responses

> 200 OK

— 302 The item was moved

— 403 Not enough privileges

— 404 Item not found

GET /thing

Response samples

200

Content type

application/json

null

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